Multi-line Equations

Long equations and alignments can be broken across multiple lines using alignment environments:

$$2x - 5y = 8 \tag{1}$$
$$3x + 9y = -12 \tag{2}$$

Piecewise Functions

Piecewise functions are built using the array environment enclosed in automatic sizing brackets:

$$f(x) = \begin{cases} -x, & x < 0 \\ 0, & x = 0 \\ x^2, & x > 0 \end{cases}$$

Matrices with Delimiters

Matrices support various enclosure types such as parentheses, brackets, or determinants:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad C = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

For inline use, compact matrices can be created using smallmatrix: $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$.